

MacroHard WeakEdge

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

I've hidden a flag in this file. Can you find it?

Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read metadata
 - `binwalk` – Detect and extract embedded files
 - `unzip` – Extract ZIP archives
 - `base64` – Decode Base64-encoded content
-

Complete Solution

Step 1: Download and Prepare

Download the file named `Forensics_is_fun.pptm` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type:

```
file Forensics_is_fun.pptm
```

Expected output:

```
Forensics_is_fun.pptm: Microsoft PowerPoint 2007+
```

The file is a **PowerPoint Macro-Enabled Presentation** (`.pptm`), which is essentially a ZIP archive containing XML files and potentially VBA macros.

Step 3: Search for Visible Text

Check if the flag appears as plain text:

```
strings Forensics_is_fun.pptm | grep -i "picoCTF"
```

Output: (no results)

The flag isn't directly visible.

Step 4: Examine Metadata

Use `exiftool` to inspect metadata:

```
exiftool Forensics_is_fun.pptm
```

Output:

```
ExifTool Version Number      : 13.50
File Name                    : Forensics_is_fun.pptm
Directory                   : .
File Size                    : 100 kB
...
File Type                   : PPTM
File Type Extension          : pptm
MIME Type                   : application/vnd.ms-
powerpoint.presentation.macroEnabled.12
Zip Required Version         : 20
Zip Bit Flag                 : 0x0006
```

```
Zip Compression      : Deflated
...
Title                : Forensics is fun
Creator              : John
Last Modified By     : John
Slides               : 58
Hidden Slides        : 1
...
```

Key finding: There is **1 hidden slide** in the presentation. This suggests the flag might be on a slide that isn't visible when normally opening the file.

Step 5: Treat the File as a ZIP Archive

Since `.pptm` files are ZIP archives, we can extract them directly:

```
unzip Forensics_is_fun.pptm
```

Output:

```
Archive:  Forensics_is_fun.pptm
  extracting: [Content_Types].xml
  extracting: _rels/.rels
  extracting: ppt/presentation.xml
  extracting: ppt/slides/_rels/slide46.xml.rels
  extracting: ppt/slides/slide1.xml
  extracting: ppt/slides/slide2.xml
  ...
  extracting: ppt/slideMasters/hidden
  ...
```

Notice the file `ppt/slideMasters/hidden` – this looks suspicious!

Step 6: Examine the Hidden File

```
cat ppt/slideMasters/hidden
```

Output:

```
Z m x h Z z o g c G l j b 0 N U R n t E M W R f d V 9 r b j B 3 X 3 B w d H N f c l 9
6 M X A 1 f Q
```

The text appears to be spaced out. Removing the spaces gives:

```
ZmxhZzogcGljb0NURntEMWRfdV9rbjB3X3BwdHNfc196MXA1fQ
```

This is a **Base64-encoded** string!

Step 7: Decode the Base64

```
echo "ZmxhZzogcGljb0NURntEMWRfdV9rbjB3X3BwdHNfc196MXA1fQ" | base64 -d
```

Output:

```
flag: picoCTF{<REDACTED>}
```

Step 8: Alternative Method – Using Binwalk

You can also use `binwalk` to detect and extract embedded files:

```
binwalk Forensics_is_fun.pptm
```

Output (excerpt):

DECIMAL	HEXADECIMAL	DESCRIPTION

0	0x0	Zip archive data, at least v2.0 to extract, ...
...		
88548	0x159E4	Zip archive data, at least v2.0 to extract, compressed size: 81, uncompressed size: 99, name: ppt/slideMasters/hidden

Then extract recursively:

```
binwalk Forensics_is_fun.pptm -Me --run-as=root
```

Then navigate to the extracted file:

```
cat _Forensics_is_fun.pptm.extracted/ppt/slideMasters/hidden
```

Same result!

Step 9: Alternative Method – Using xxd to Verify ZIP Header

```
xxd Forensics_is_fun.pptm | head
```

Output:

```
00000000: 504b 0304 1400 0600 0800 0000 2100 977e  PK.....!...~
00000010: 51a0 a202 0000 a429 0000 1300 0802 5b43  Q.....).....[C
00000020: 6f6e 7465 6e74 5f54 7970 6573 5d2e 786d  ontent_Types].xm
...
```

The `PK` signature confirms this is a valid ZIP archive.



Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, decoding the Base64 string will reveal the complete flag.



Key Concepts

Concept	Description
PPTM Format	PowerPoint Macro-Enabled Presentation – a ZIP archive containing XML files, media, and VBA macros.

Concept	Description
ZIP Archive	A compressed file format that can contain multiple files and folders.
Hidden Slides	PowerPoint slides that are marked as hidden and don't appear in normal slideshow mode.
Base64 Encoding	A common encoding used to represent binary data in ASCII text (detected by <code>=</code> padding or alphanumeric patterns).
binwalk	A tool for detecting and extracting embedded files from within other files.
File Extension Trust	Never rely solely on file extensions; use <code>file</code> and <code>exiftool</code> to verify actual content.

Pro Tips

1. **Office documents are ZIP files** – `.docx`, `.xlsx`, `.pptx`, and `.pptm` files are actually ZIP archives. Rename them to `.zip` or use `unzip` directly.
2. **Hidden slides can hide flags** – Always check for hidden content in presentation files.
3. **Spaced-out text is often Base64** – When you see text with spaces between characters, removing the spaces often reveals a Base64 string.
4. **exiftool reveals metadata** – It showed "Hidden Slides: 1", which was the key clue.
5. **Use binwalk for deep inspection** – It can detect embedded files even when the file structure is corrupted.
6. **unzip is simpler for Office documents** – Since these are standard ZIP files, `unzip` works perfectly without needing `binwalk`.

Alternative Base64 Decoding Tools

Tool	Command / Link
Linux <code>base64</code>	<code>echo "string" base64 -d</code>
CyberChef	https://gchq.github.io/CyberChef/ (Operation: From Base64)
Python	<code>import base64; base64.b64decode("string")</code>
Online	https://www.base64decode.org/